Présentation Projet SMACP

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Willan BANOR
Eliott MOLARD

Sommaire

Contexte / Problématique

Etude du Laboratoire PIMENT Université de la Réunion

Fonctionnel de 2019 - 2024

Implémentation par le SIDELEC dans le cirque de Mafate

Répartition des tâches

Étudiant 1 : Mathias DIJOUX

- Acquisition des données électriques des panneaux photovoltaïque
- Acquisition des données de rayonnement solaire et données environnementales in situ
- Déploiement de la solution Arduino Wifi
- Détection de défauts et mise en protection
- Cybersécurisation et tableau de bord de la partie production électrique

Étudiant 2 : Willan BANOR

- Acquisition des données du pack de batterie et du contrôle de la charge
- Cartographie GPS pour la position de la centrale
- Déploiement de la solution type bus de terrain (CAN ou RS485)
- Détection des défauts et mise en protection
- Cybersécurisation et tableau de bord de la partie stockage

Étudiant 3 : Eliott MOLARD

- Tableau de bord administrateur
- Infrastructure réseau LAN / IoT
- Intégration des solutions E1 ,E2
- Cybersécurisation de la solution

Objectif du projet

- Besoin de supervision à distance
- Centraliser les données
- Sécuriser l'installation
- Surveiller les performances

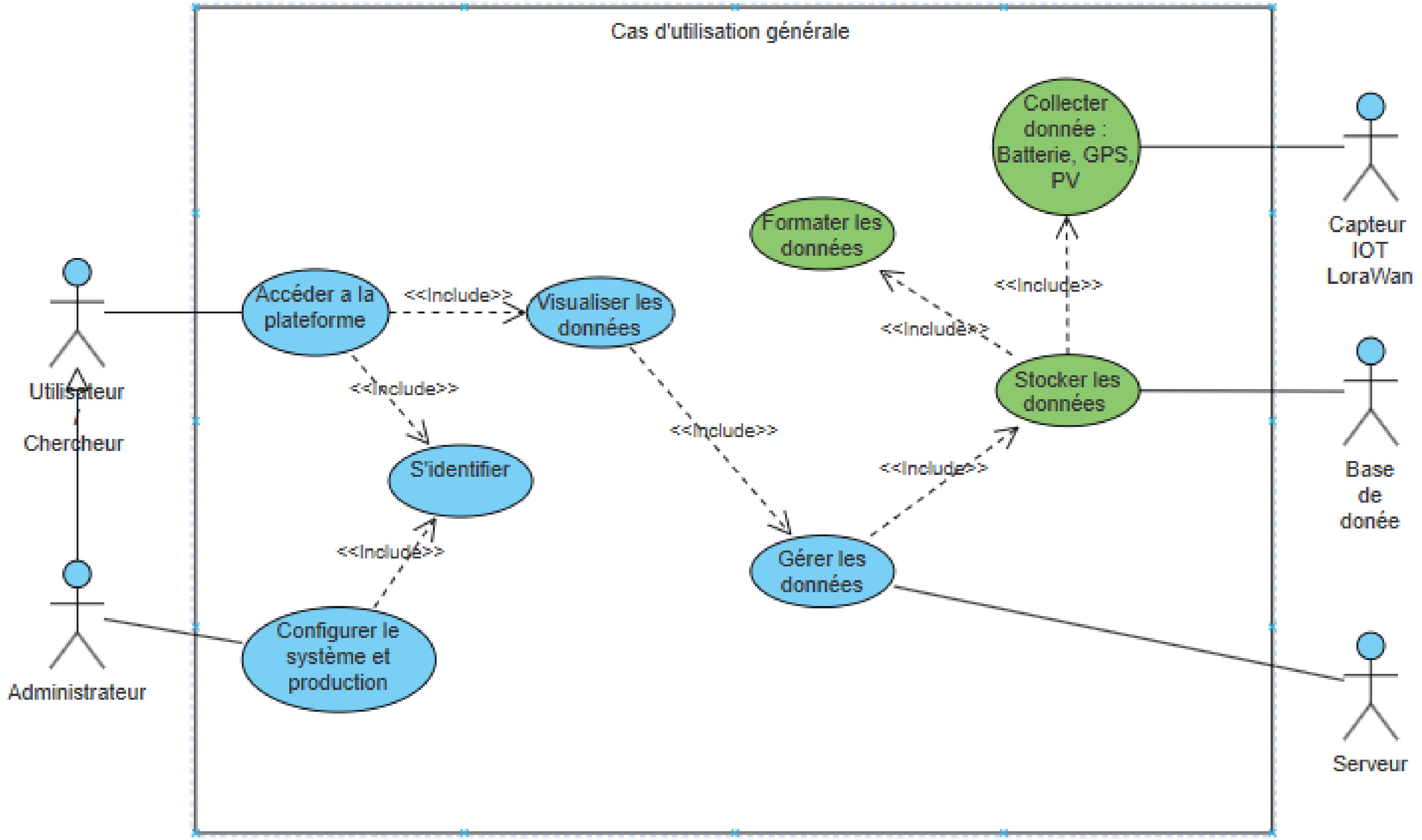
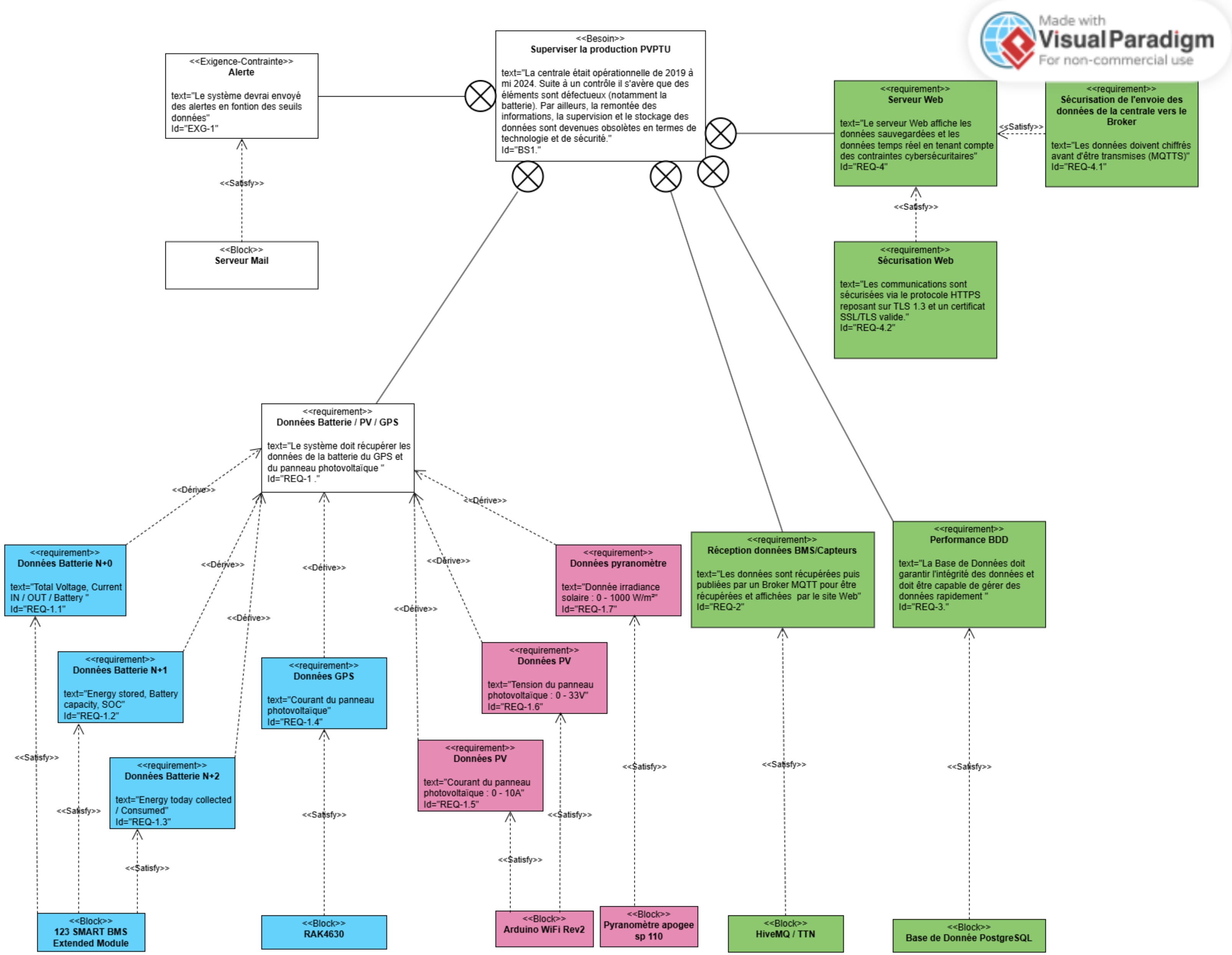


Diagramme d’éxigence



Synoptique projet

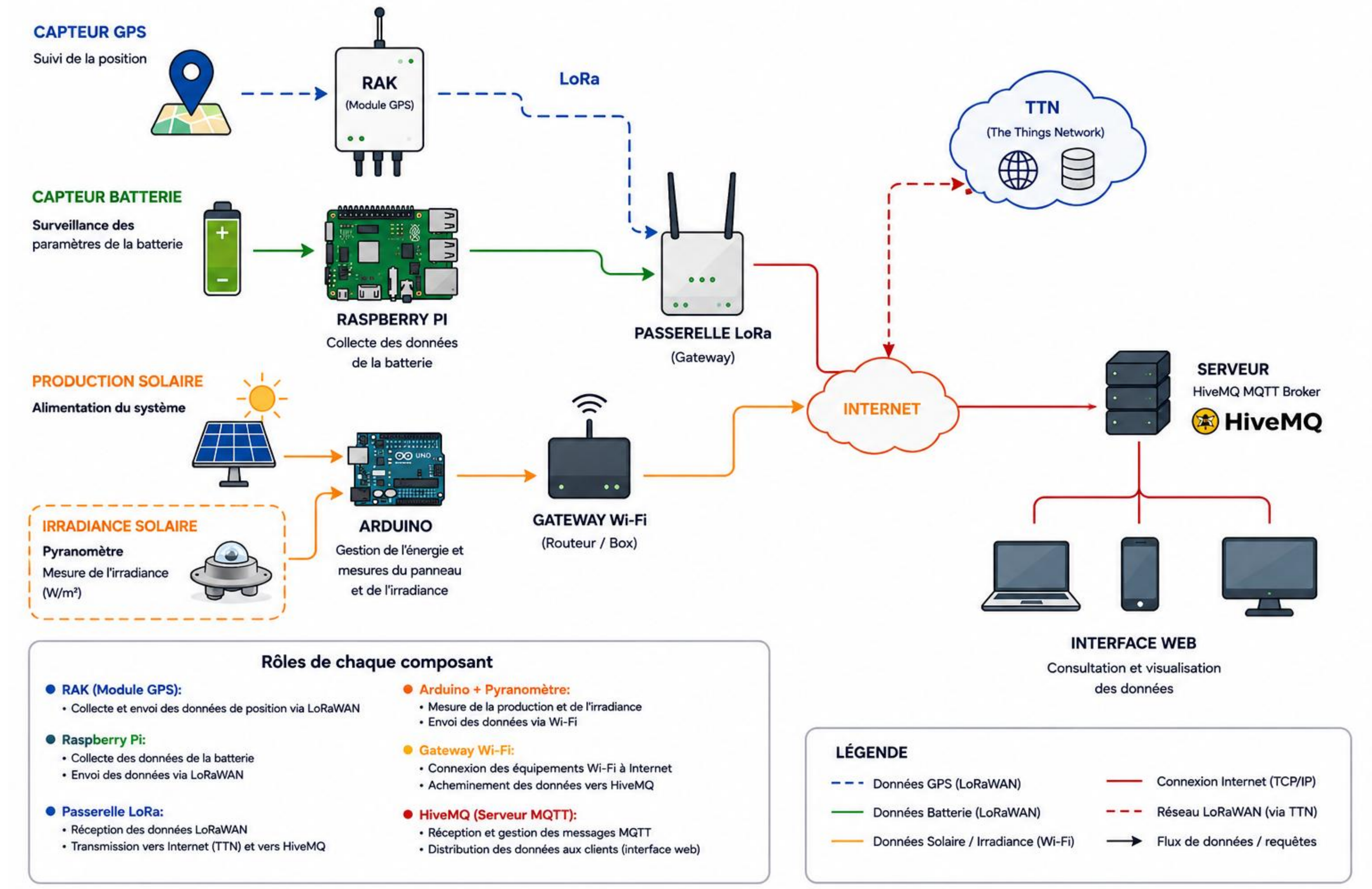


Diagramme de Cas d’utilisation (personnel)

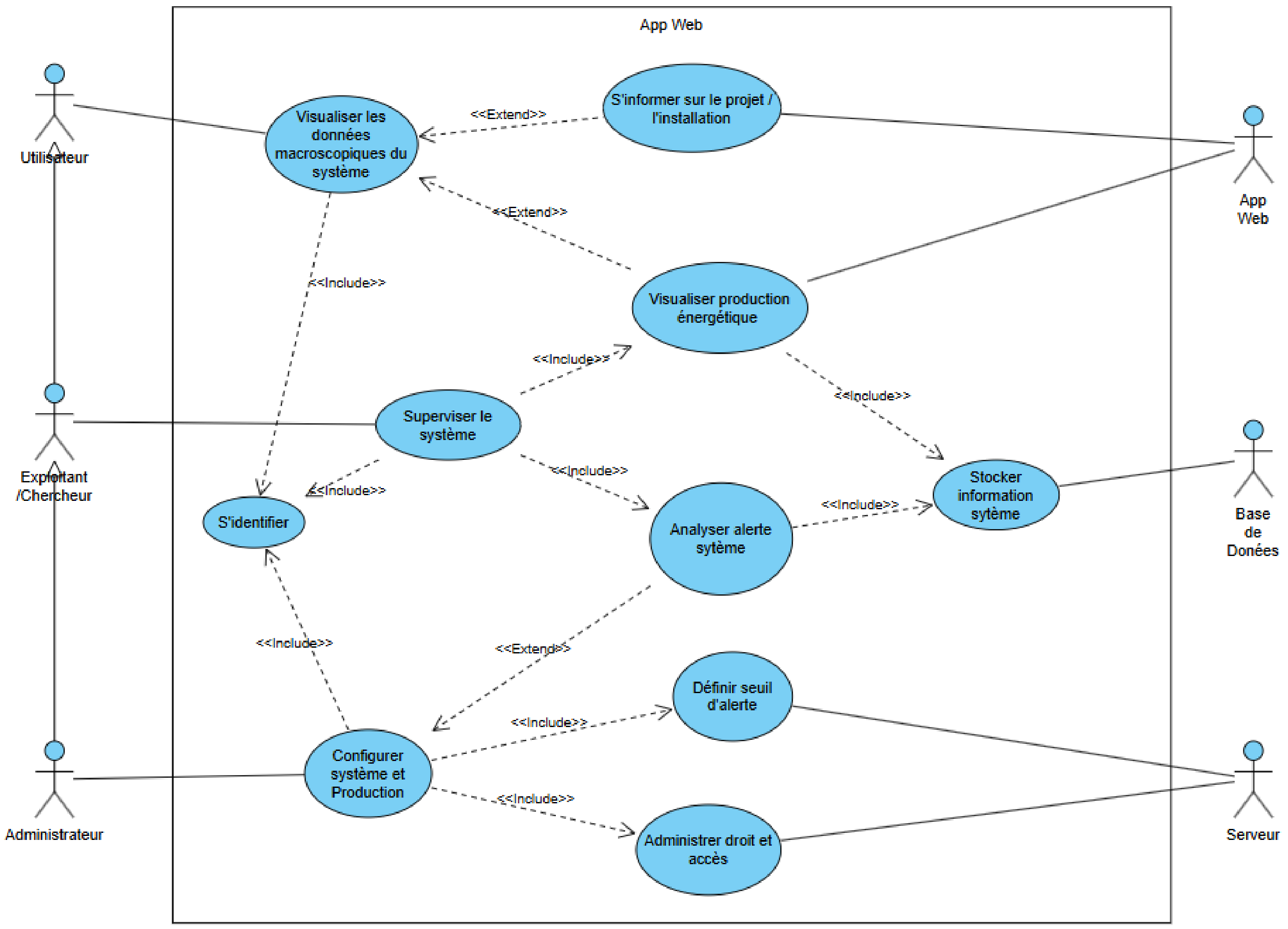
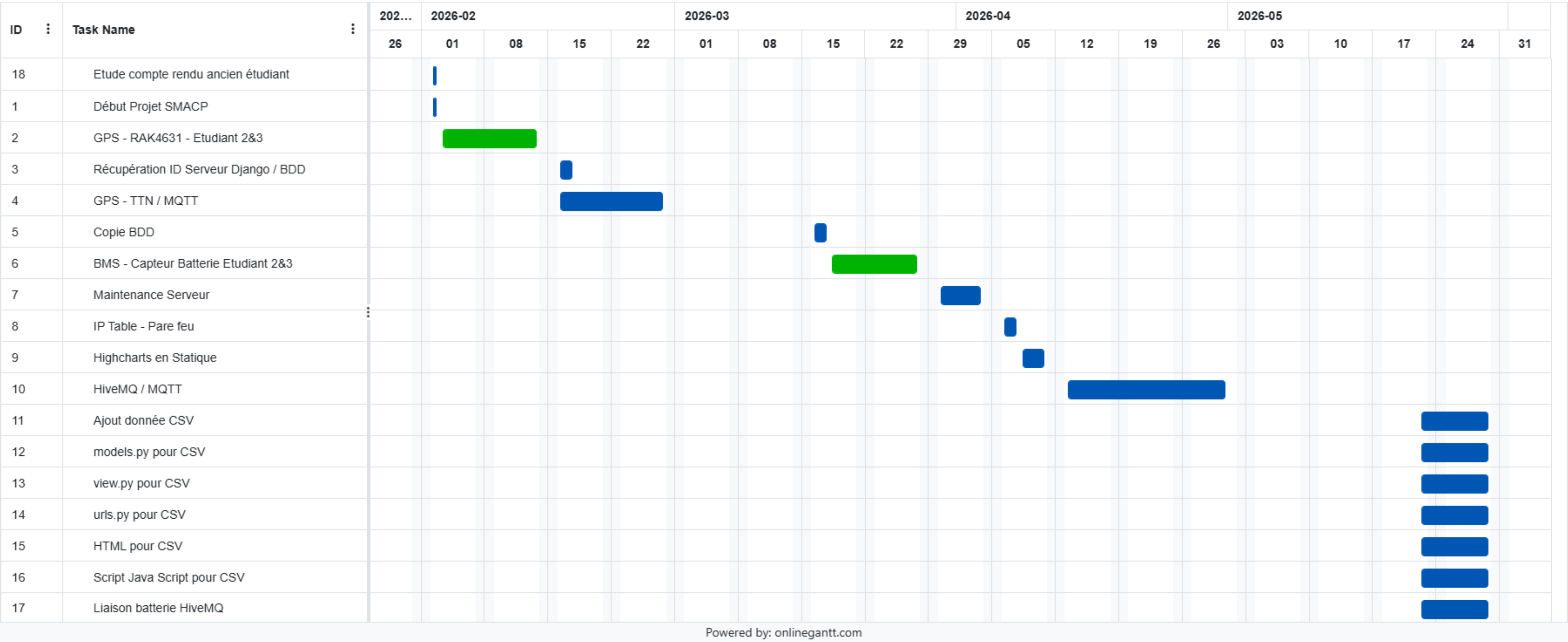
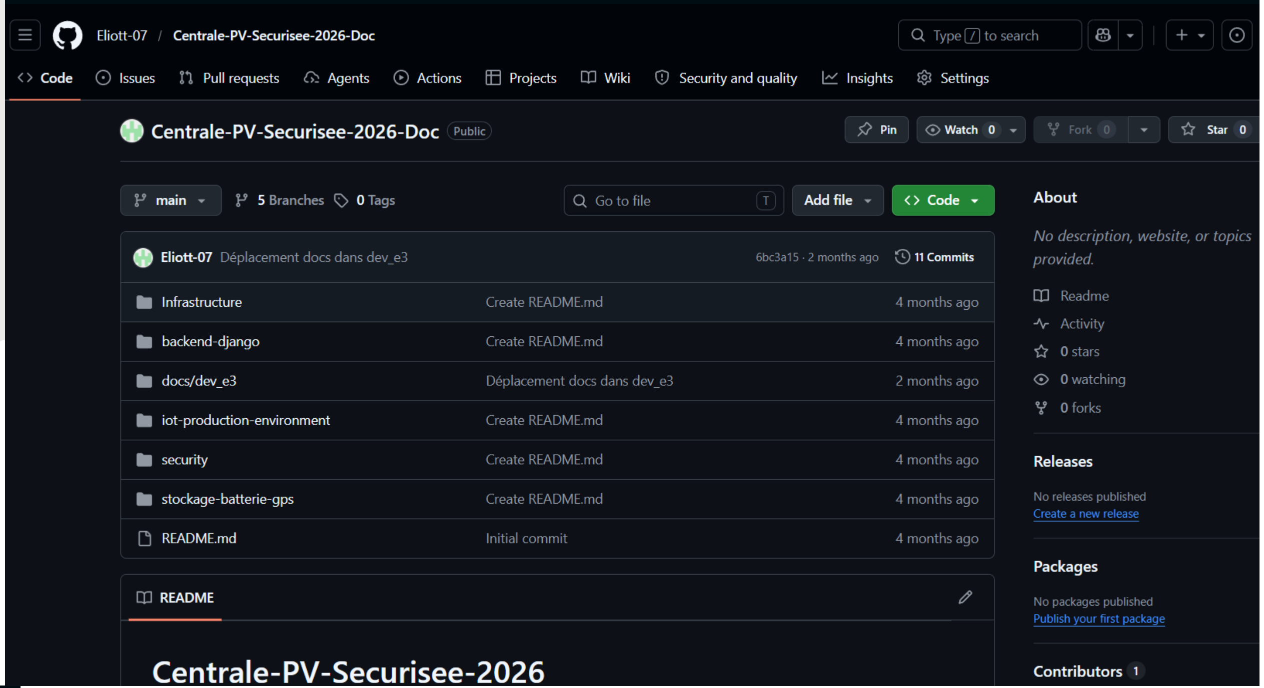
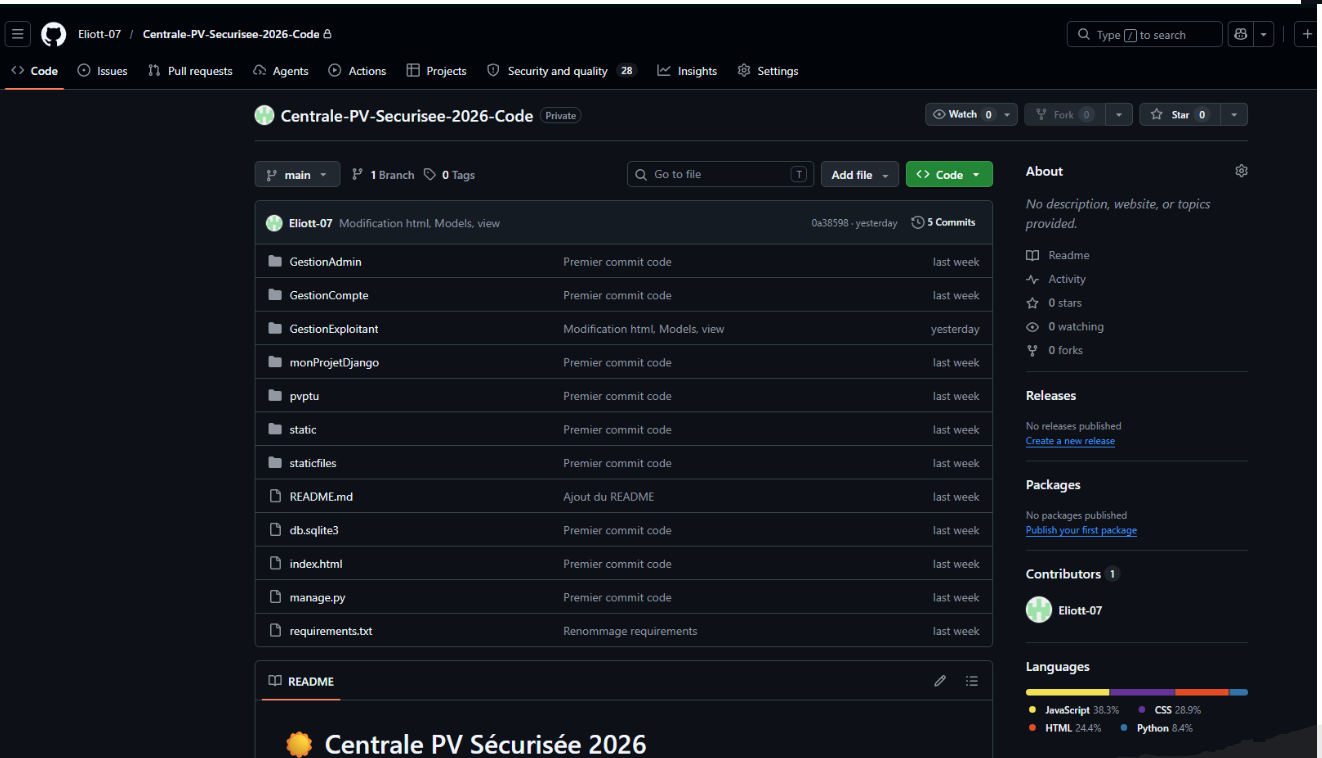


Diagramme de Gantt



GitHub





2 Dépôt :

- 1 pour la documentation
- 1 pour le code

Objectifs du Projet (partie personnel)

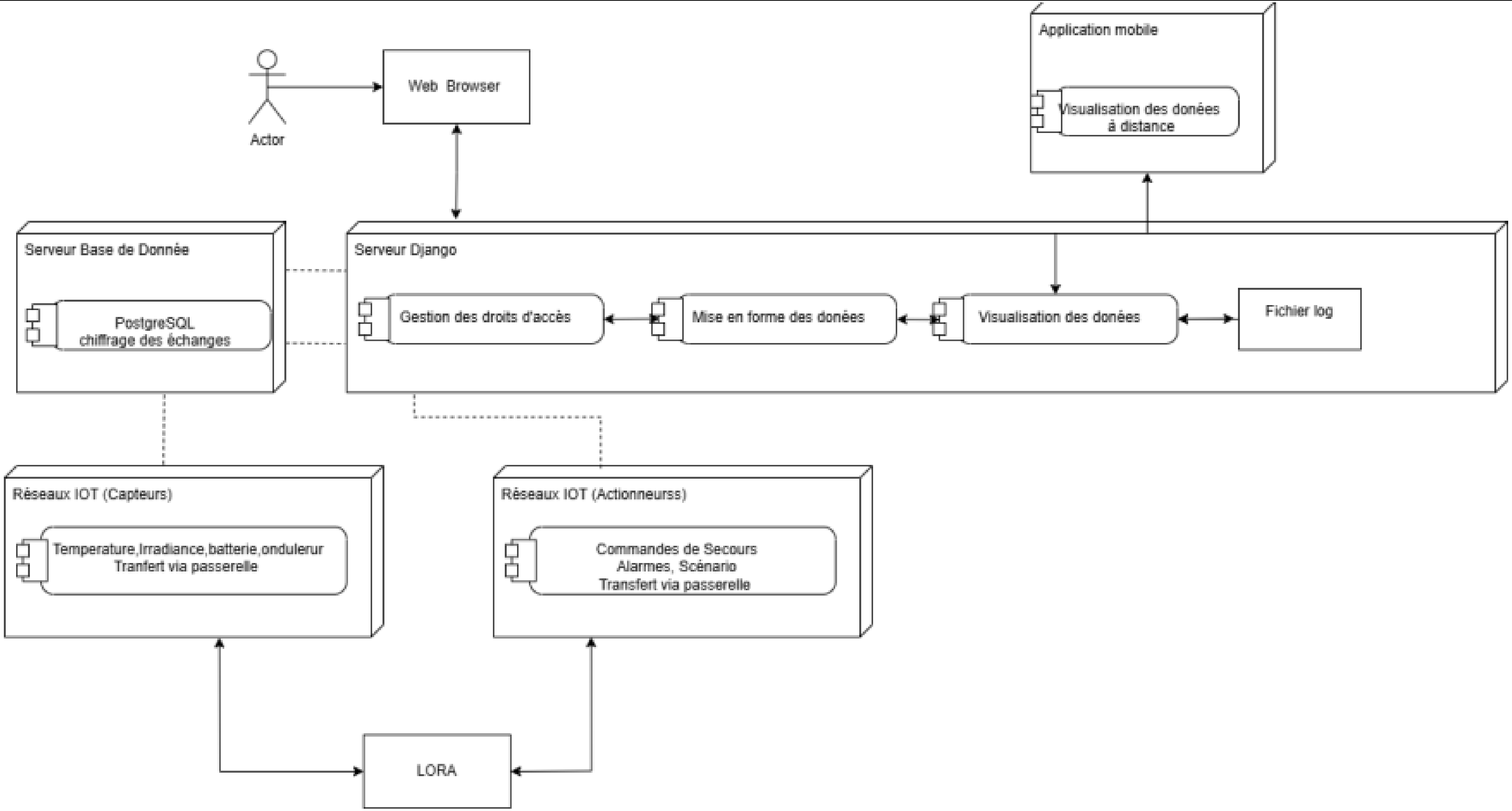
Intégrer les nouvelles fonctionnalités de mes collègues 2026

Intégrer toutes les données N0 à N7 (actuellement seulement N6)

Vérification des règles de firewall

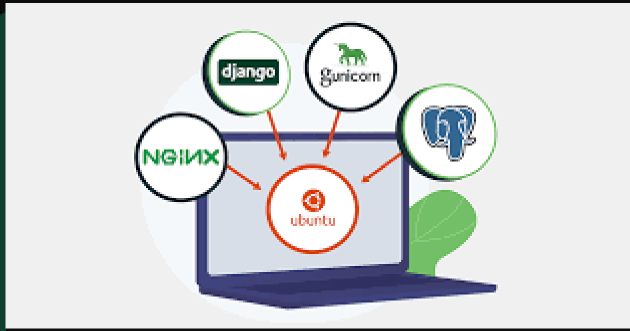
Correction des codes

Architecture Générale



Technologies utilisées

- Python
- Django
- PostgreSQL
- MQTT
- HiveMQ
- Ubuntu 24.10
- Nginx
- Gunicorn
- Vs Code



Pourquoi Django ?

Rapide a mettre en place

Très complet

Sécurisé

Evolutif et Polyvalent

Fonctionnement de l'application

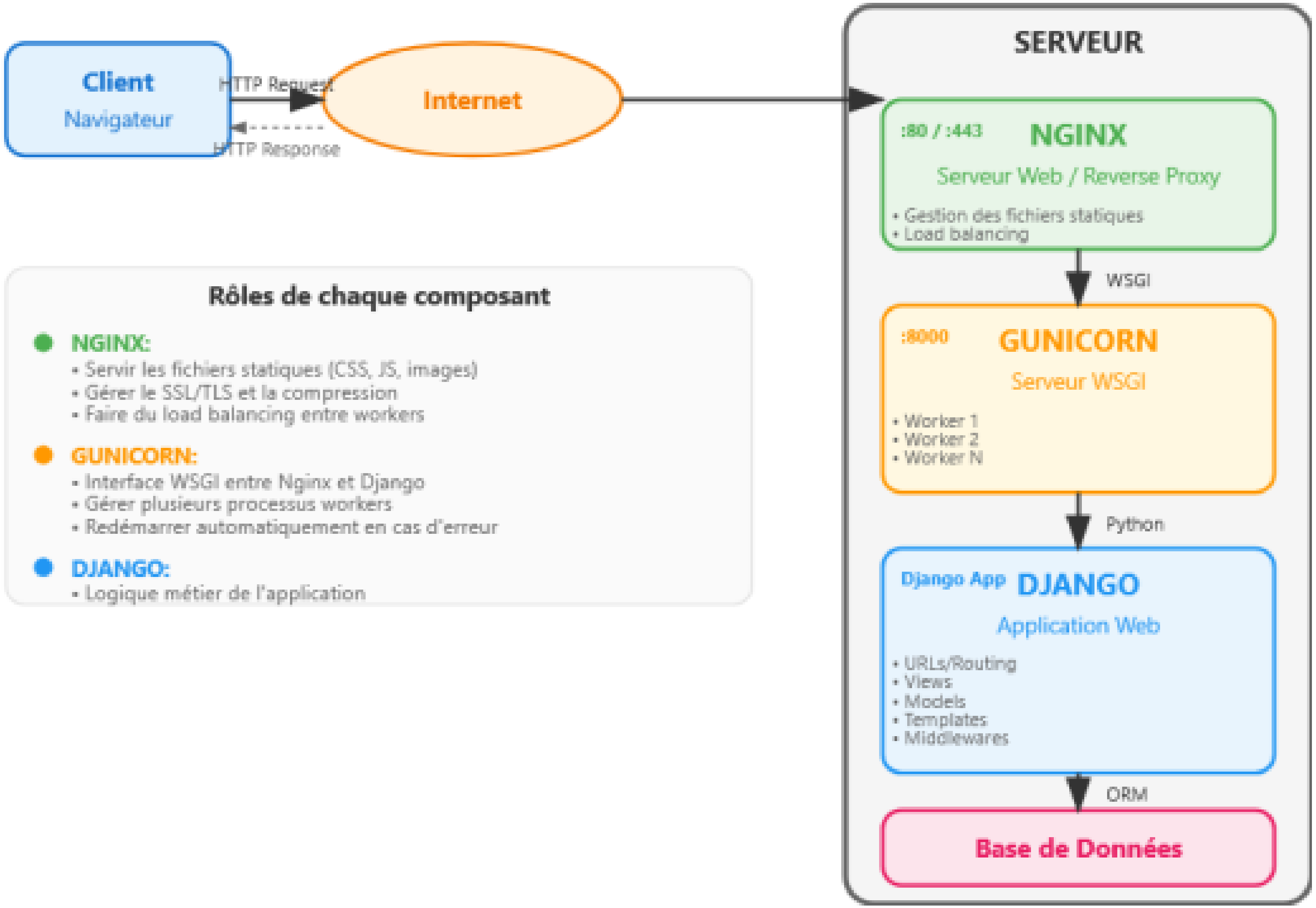
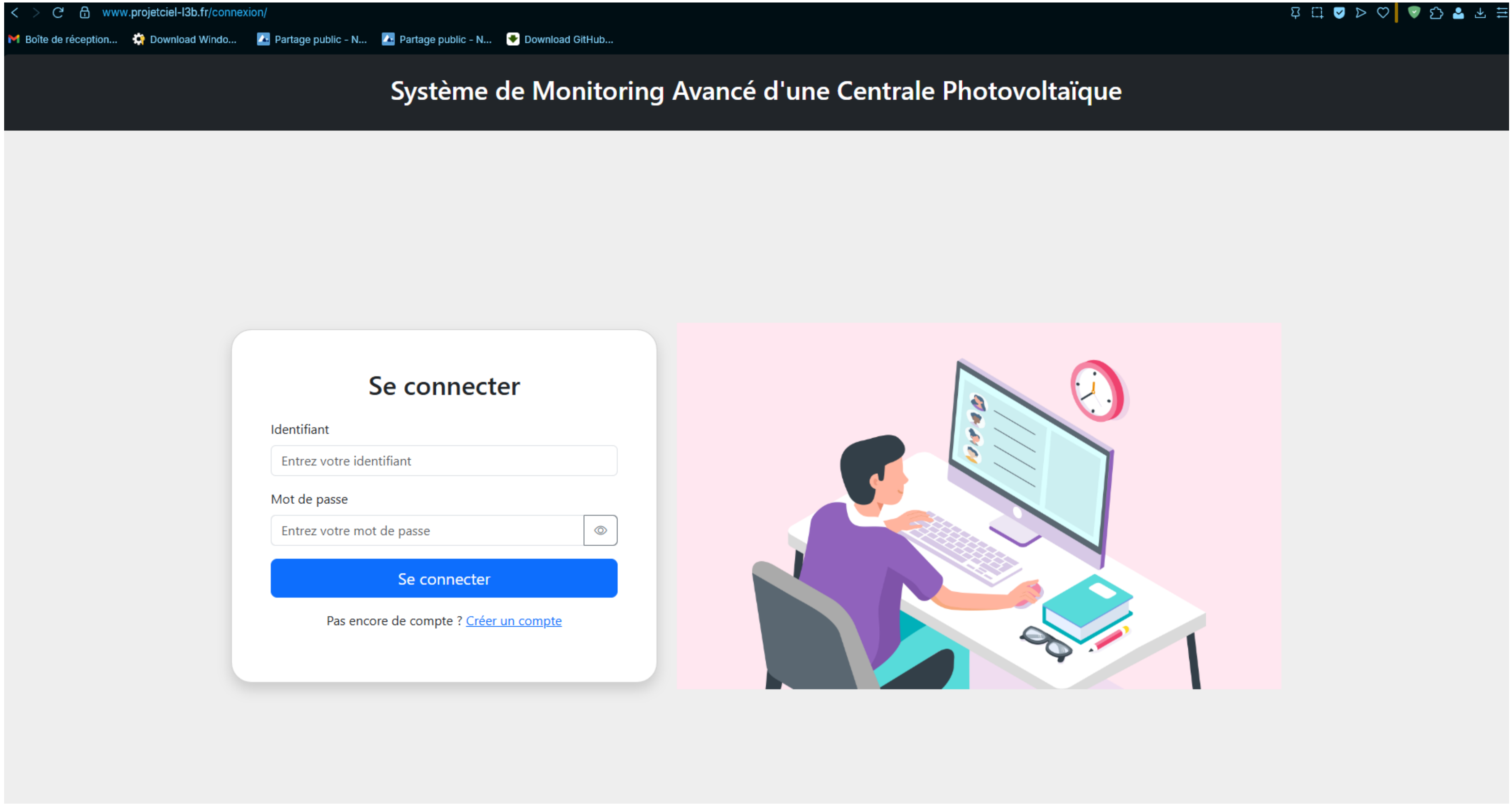


Schéma structurel de la communication

Interface Utilisateur

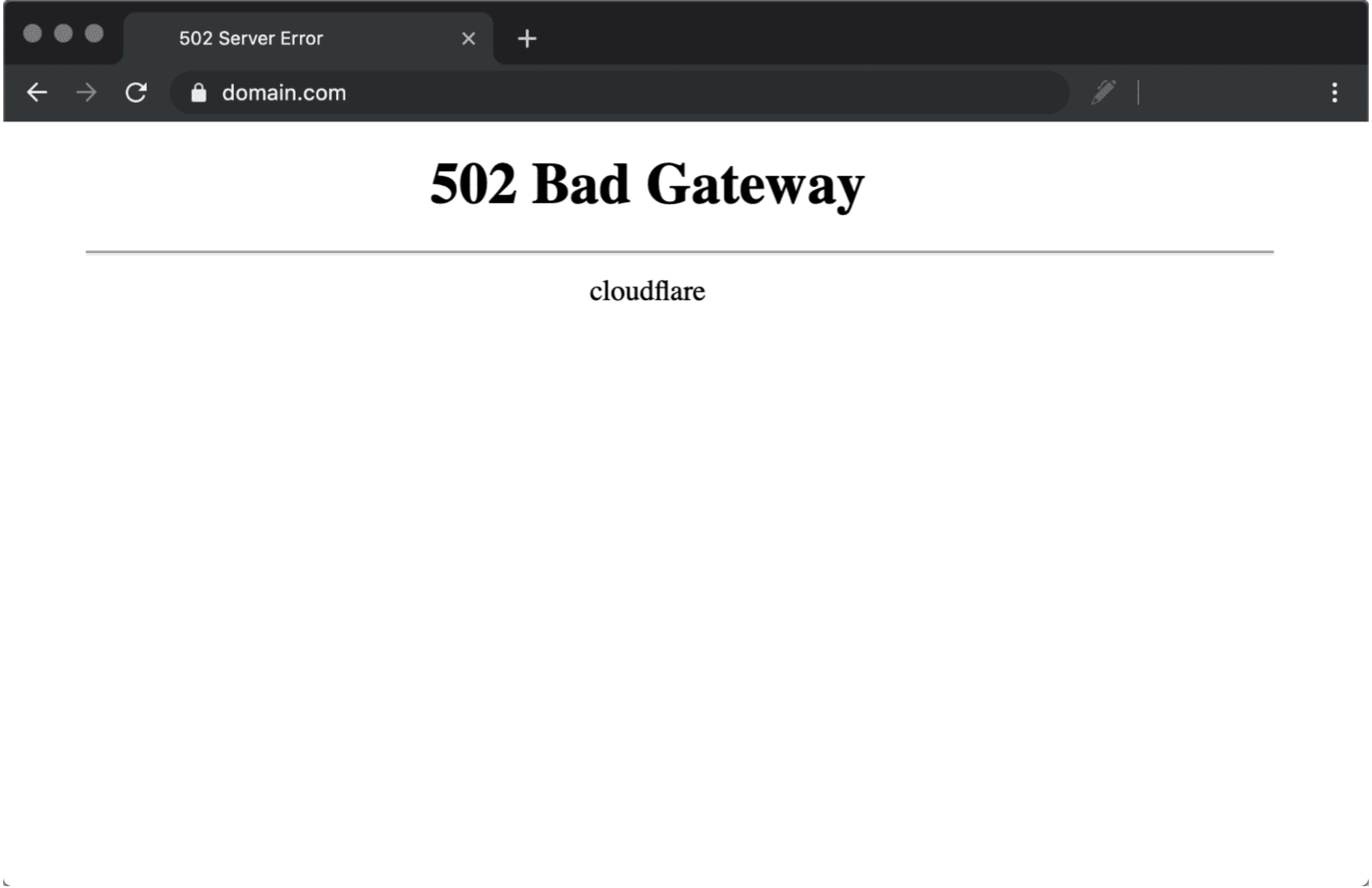


Page d’accueil du site

Difficultés rencontrées / Correctifs & Travail réalisé

- Documentation incomplete / peu détaillée
- Git push défectueux
- Liens Highcharts invalides
- Gunicorn
- Intégration donée Batterie → Django
- Liason Batterie → HiveMQ → Django

```
ubuntu@vps-62f78ac3:~$ sudo systemctl status gunicorn
× gunicorn.service - gunicorn daemon
     Loaded: loaded (/etc/systemd/system/gunicorn.service; enabled; preset: enabled)
     Active: failed (Result: exit-code) since Wed 2026-04-01 19:32:32 UTC; 14h ago
   Duration: 135ms
 Invocation: 18a714d499f54382bcf817927be0a491
TriggeredBy: × gunicorn.socket
    Process: 1385 ExecStart=/home/ubuntu/dev/envdjango/bin/gunicorn --access-logfile - --workers 3 --bind unix:/run/gunicorn.sock monProjetDjango.wsgi:application (code=exited, status=3)
   Main PID: 1385 (code=exited, status=3)

Apr 01 19:32:31 vps-62f78ac3 gunicorn[1386]: ModuleNotFoundError: No module named 'monProjetDjango.wsgi'
Apr 01 19:32:31 vps-62f78ac3 gunicorn[1386]: [2026-04-01 19:32:31 +0000] [1386] [INFO] Worker exiting (pid: 1386)
Apr 01 19:32:31 vps-62f78ac3 gunicorn[1385]: [2026-04-01 19:32:31 +0000] [1385] [ERROR] Worker (pid:1386) exited with code 3
Apr 01 19:32:31 vps-62f78ac3 gunicorn[1385]: [2026-04-01 19:32:31 +0000] [1385] [ERROR] Shutting down: Master
Apr 01 19:32:31 vps-62f78ac3 gunicorn[1385]: [2026-04-01 19:32:31 +0000] [1385] [ERROR] Reason: Worker failed to boot.
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: gunicorn.service: Main process exited, code=exited, status=3/NOTIMPLEMENTED
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: gunicorn.service: Failed with result 'exit-code'.
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: gunicorn.service: Start request repeated too quickly.
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: gunicorn.service: Failed with result 'exit-code'.
Apr 01 19:32:32 vps-62f78ac3 systemd[1]: Failed to start gunicorn.service - gunicorn daemon.
ubuntu@vps-62f78ac3:~$
```

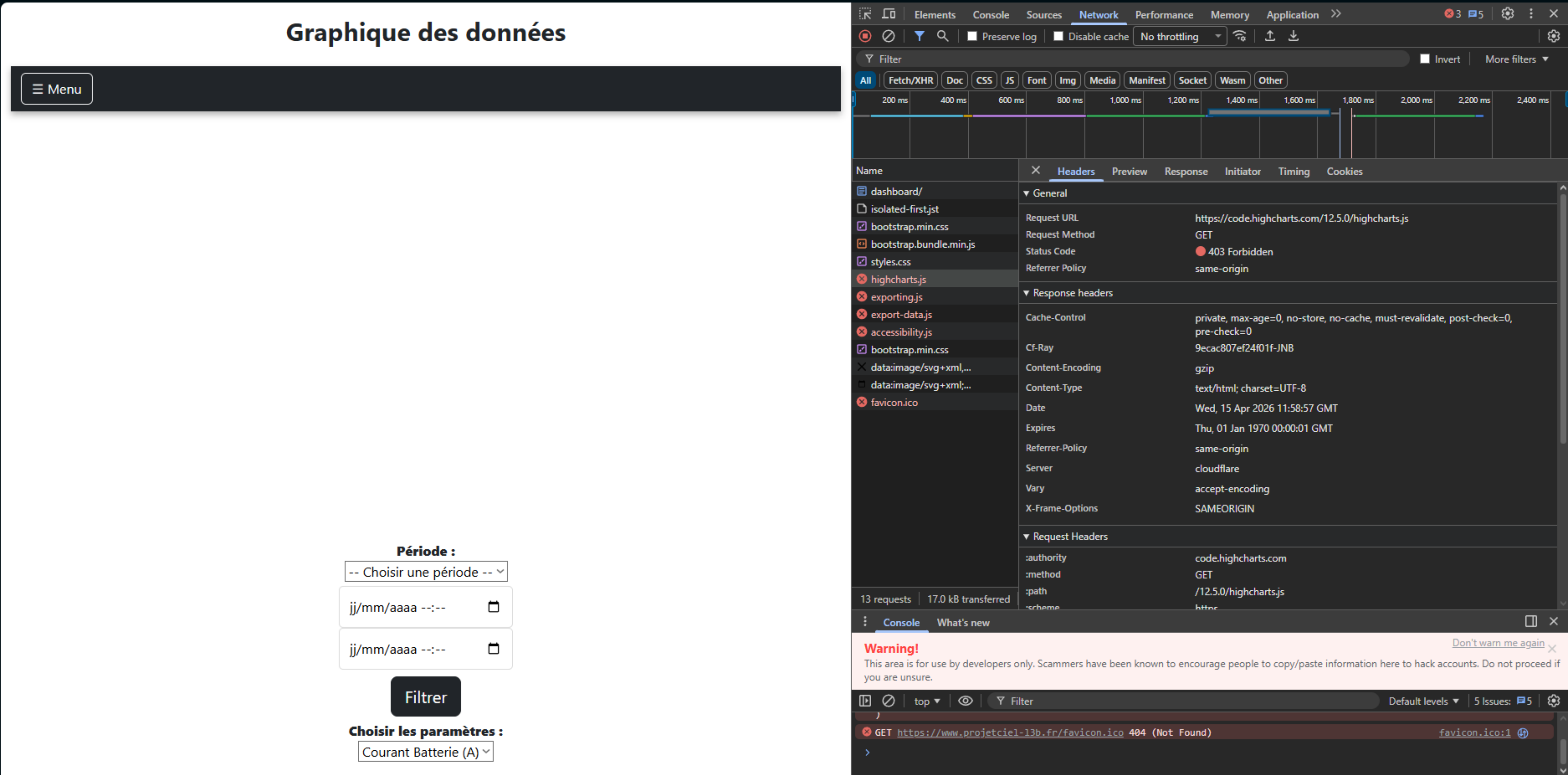

502 Server Error
domain.com
502 Bad Gateway
cloudflare

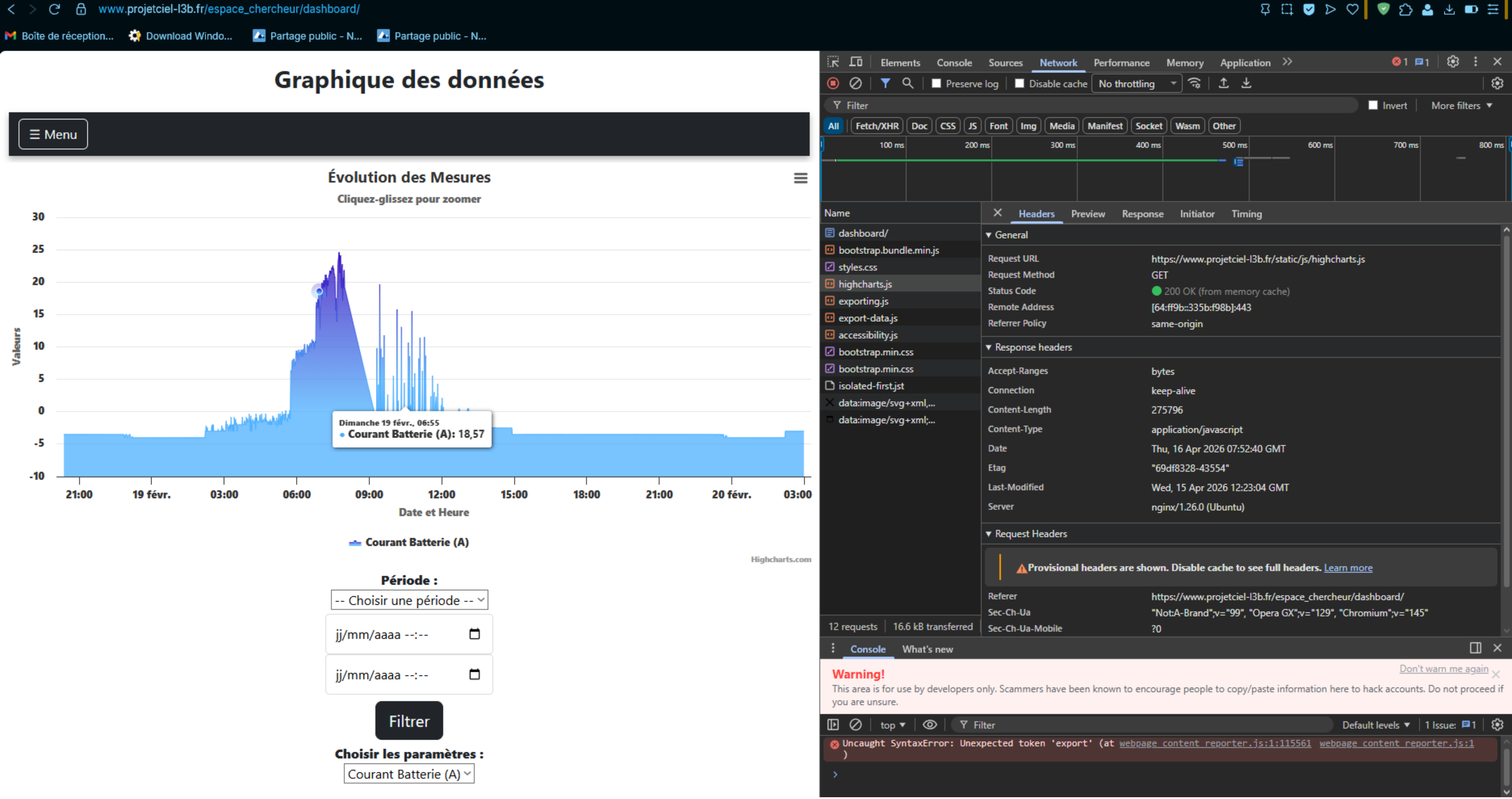
```html
    <meta charset="UTF-8">
    <title>Dashboard</title>
    <link rel="stylesheet" href="{% static 'css/styles.css' %}">
    <script src="https://code.highcharts.com/highcharts.js"></script>
    <script src="https://code.highcharts.com/modules/exporting.js"></script>
    <script src="https://code.highcharts.com/modules/export-data.js"></script>
    <script src="https://code.highcharts.com/modules/accessibility.js"></script>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
```

Liens CDN Highcharts

```html
<head>
    <meta charset="UTF-8">
    <title>Dashboard</title>
    <link rel="stylesheet" href="{% static 'css/styles.css' %}">
    <script src="{% static 'js/highcharts.js' %}"></script>
    <script src="{% static 'js/exporting.js' %}"></script>
    <script src="{% static 'js/export-data.js' %}"></script>
    <script src="{% static 'js/accessibility.js' %}"></script>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css" rel="stylesheet">
</head>
```

Liens statique Highcharts


Graphique des données
☰ Menu
Période :
-- Choisir une période --
jj/mm/aaaa --:--
jj/mm/aaaa --:--
Filtrer
Choisir les paramètres :
Courant Batterie (A)
Elements Console Sources Network Performance Memory Application
Preserve log Disable cache No throttling
Filter Invert More filters
All Fetch/XHR Doc CSS JS Font Img Media Manifest Socket Wasm Other
Name
dashboard/
isolated-first.jst
bootstrap.min.css
bootstrap.bundle.min.js
styles.css
highcharts.js
exporting.js
export-data.js
accessibility.js
bootstrap.min.css
data:image/svg+xml,...
data:image/svg+xml;...
favicon.ico
13 requests 17.0 kB transferred
Headers Preview Response Initiator Timing Cookies
General
Request URL https://code.highcharts.com/12.5.0/highcharts.js
Request Method GET
Status Code 403 Forbidden
Referrer Policy same-origin
Response headers
Cache-Control private, max-age=0, no-store, no-cache, must-revalidate, post-check=0, pre-check=0
Cf-Ray 9ecac807ef24f01f-JNB
Content-Encoding gzip
Content-Type text/html; charset=UTF-8
Date Wed, 15 Apr 2026 11:58:57 GMT
Expires Thu, 01 Jan 1970 00:00:01 GMT
Referrer-Policy same-origin
Server cloudflare
Vary accept-encoding
X-Frame-Options SAMEORIGIN
Request Headers
:authority code.highcharts.com
:method GET
:path /12.5.0/highcharts.js
Console What's new
Warning!
Don't warn me again
This area is for use by developers only. Scammers have been known to encourage people to copy/paste information here to hack accounts. Do not proceed if you are unsure.
top Filter Default levels 5 Issues: 5
GET https://www.projetciel-13b.fr/favicon.ico 404 (Not Found) favicon.ico:1



www.projetciel-l3b.fr/espace_chercheur/dashboard/
Boîte de réception...
Download Windo...
Partage public - N...
Partage public - N...
Graphique des données
☰ Menu
Évolution des Mesures
Cliquez-glissez pour zoomer
Valeurs
30
25
20
15
10
5
0
-5
-10
21:00
19 févr.
03:00
06:00
09:00
12:00
15:00
18:00
21:00
20 févr.
03:00
Dimanche 19 févr., 06:55
Courant Batterie (A): 18,57
Date et Heure
Courant Batterie (A)
Highcharts.com
Période :
-- Choisir une période --
jj/mm/aaaa --:--
jj/mm/aaaa --:--
Filtrer
Choisir les paramètres :
Courant Batterie (A)
Elements
Console
Sources
Network
Performance
Memory
Application
Preserve log
Disable cache
No throttling
Filter
Invert
More filters
All
Fetch/XHR
Doc
CSS
JS
Font
Img
Media
Manifest
Socket
Wasm
Other
100 ms
200 ms
300 ms
400 ms
500 ms
600 ms
700 ms
800 ms
Name
dashboard/
bootstrap.bundle.min.js
styles.css
highcharts.js
exporting.js
export-data.js
accessibility.js
bootstrap.min.css
bootstrap.min.css
isolated-first.jst
data:image/svg+xml,...
data:image/svg+xml;...
12 requests
16.6 kB transferred
Headers
Preview
Response
Initiator
Timing
General
Request URL
https://www.projetciel-l3b.fr/static/js/highcharts.js
Request Method
GET
Status Code
200 OK (from memory cache)
Remote Address
[64:ff9b::335b:f98b]:443
Referrer Policy
same-origin
Response headers
Accept-Ranges
bytes
Connection
keep-alive
Content-Length
275796
Content-Type
application/javascript
Date
Thu, 16 Apr 2026 07:52:40 GMT
Etag
"69df8328-43554"
Last-Modified
Wed, 15 Apr 2026 12:23:04 GMT
Server
nginx/1.26.0 (Ubuntu)
Request Headers
Provisional headers are shown. Disable cache to see full headers. Learn more
Referer
https://www.projetciel-l3b.fr/espace_chercheur/dashboard/
Sec-Ch-Ua
"Not:A-Brand";v="99", "Opera GX";v="129", "Chromium";v="145"
Sec-Ch-Ua-Mobile
?0
Console
What's new
Warning!
Don't warn me again
This area is for use by developers only. Scammers have been known to encourage people to copy/paste information here to hack accounts. Do not proceed if you are unsure.
top
Filter
Default levels
1 Issue: 1
Uncaught SyntaxError: Unexpected token 'export' (at webpage_content_reporter.js:1:115561 webpage_content_reporter.js:1
)

Integration données Batterie → Django

<table>
<tr><th>Address:</th><th>Byte 0</th><th>Byte 1</th><th>Byte 2</th><th>Byte 3</th><th>Byte 4</th><th>Byte 5</th><th>Byte 6</th><th>Byte 7</th></tr>
<tr><td>N+0</td><td colspan="2">Total voltage</td><td colspan="2">Current IN</td><td colspan="2">Current OUT</td><td colspan="2">Current Battery</td></tr>
<tr><td>N+1</td><td colspan="4">Energy stored</td><td colspan="2">Battery capacity</td><td>SOC</td><td>-</td></tr>
<tr><td>N+2</td><td colspan="4">Energy today collected</td><td colspan="4">Energy today consumed</td></tr>
<tr><td>N+3</td><td colspan="4">Total energy collected</td><td colspan="4">Total energy consumed</td></tr>
<tr><td>N+4</td><td colspan="2">Cell voltage MIN</td><td colspan="2">Cell voltage MAX</td><td>Cell voltage Bypass</td><td></td><td>-</td><td>-</td></tr>
<tr><td>N+5</td><td colspan="2">Cell voltage Lowest</td><td>Low Nr</td><td colspan="2">Cell voltage Highest</td><td>High Nr</td><td>Sbyte 1</td><td>Sbyte2</td></tr>
<tr><td>N+6</td><td>Tmp,Lowest</td><td>Low Nr</td><td>Tmp,Highest</td><td>High Nr</td><td>Min charg temp</td><td>Min dis temp</td><td>Max temp</td><td>-</td></tr>
<tr><td>N+7</td><td colspan="2">Current cell voltage</td><td>Cur.Temp</td><td>Cur. Nr</td><td>Cell cnt</td><td>-</td><td>Isolation resistance</td><td></td></tr>
</table>

Données CSV → Django

- Models.py
- View.py
- Urls.py
- Fichier HTML
- Fichier JavaScript

```
class Donnee(models.Model):
    date = models.DateTimeField(primary_key=True)
    total_volt = models.FloatField()
    currentin = models.FloatField()
    currentout = models.FloatField()
    current_bat = models.FloatField()

    class Meta:
        managed = False
        db_table = 'current'

class Nzero(models.Model):
    date = models.DateTimeField(primary_key = True)
    total_volt = models.FloatField()
    currentin = models.FloatField()
    currentout = models.FloatField()
    current_bat = models.FloatField()
    class Meta:
        managed = False
        db_table = 'DB_nzero'
```

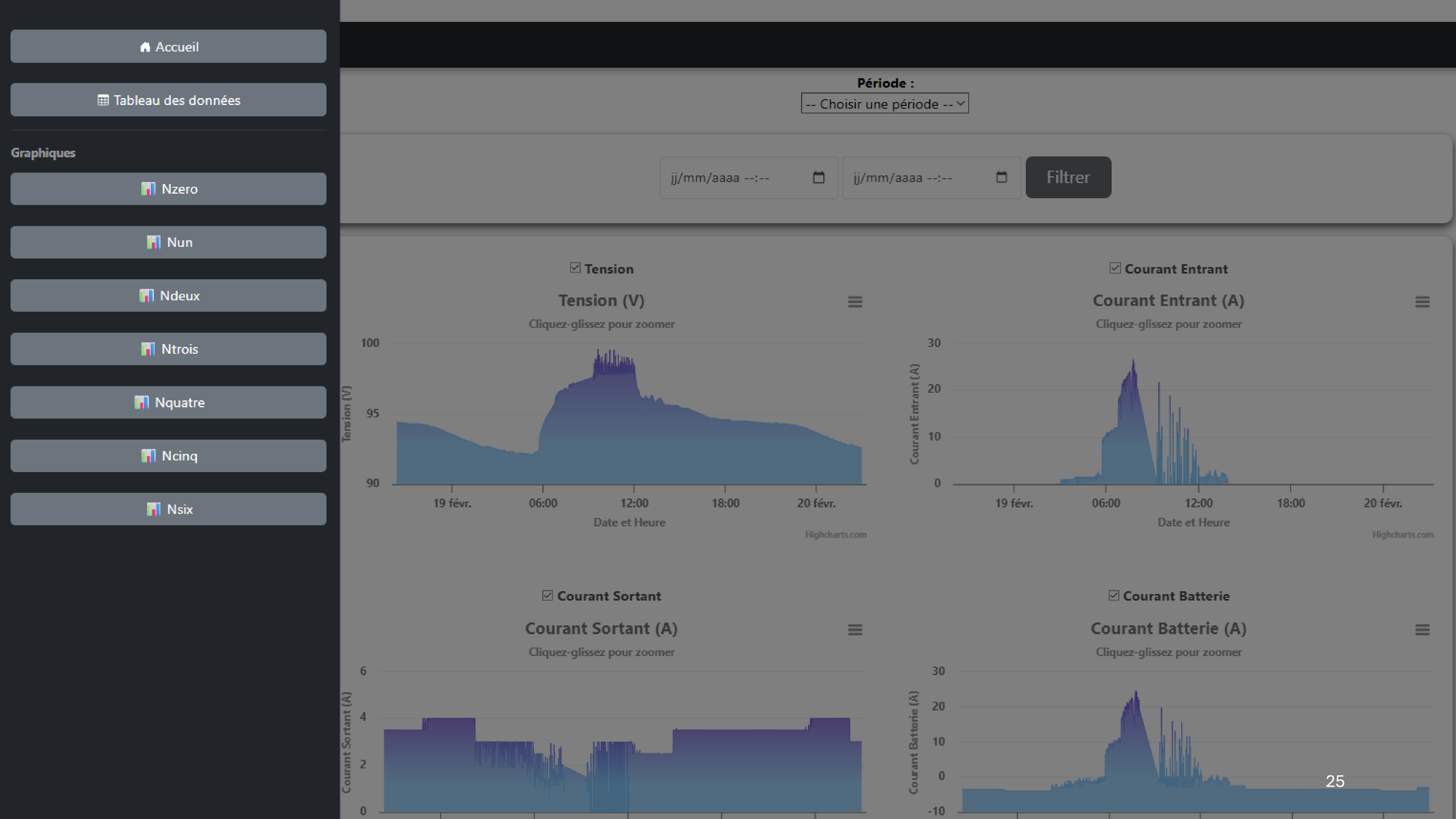
Models.py

```
    #ajout 2026
    path('grapheDesDonnees_nun/', views.GrapheDesDonnees_nun, name='visualisation_graphique_nun'),
    path('grapheDesDonnees_ndeux/',views.GrapheDesDonnees_ndeux, name='visualisation_graphique_ndeux'),
    path('grapheDesDonnees_ntrois/', views.GrapheDesDonnees_ntrois, name='visualisation_graphique_ntrois'),
    path('grapheDesDonnees_nquatre/', views.GrapheDesDonnees_nquatre, name='visualisation_graphique_nquatre'),
    path('grapheDesDonnees_ncinq/', views.GrapheDesDonnees_ncinq, name='visualisation_graphique_ncinq'),
    path('grapheDesDonnees_nsix/',views.GrapheDesDonnees_nsix, name='visualisation_graphique_nsix'),
]
```

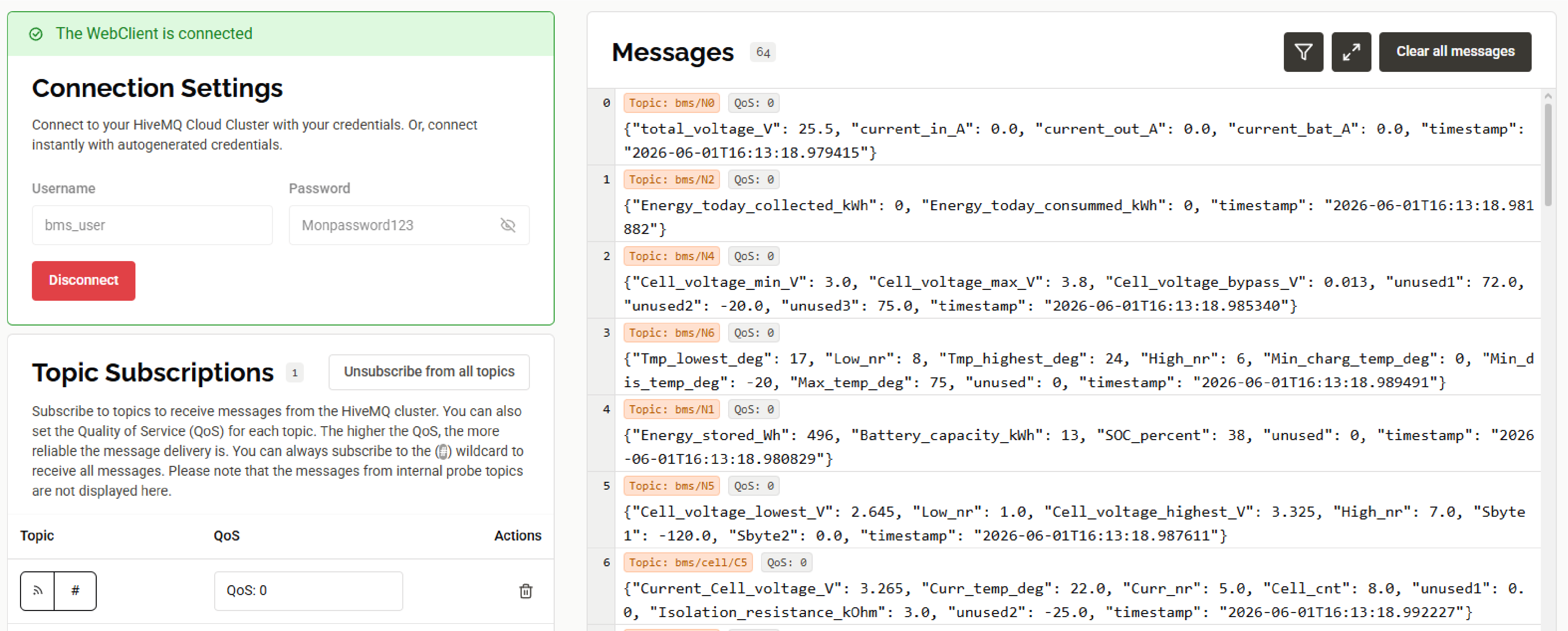
Urls.py

```
def GrapheDesDonnees(request):
    data = list(Donnee.objects.values('date', 'total_volt', 'currentin', 'currentout', 'current_bat'))
    data_json = json.dumps(data, default=str)  # Convertir QuerySet en JSON
    return render(request, "visualisation_graphique.html", {'data_json': data_json})

def GrapheDesDonnees_nun(request):
    data = list(Nun.objects.values('date', 'Energy_stored_Wh', 'Battery_capacity_kWh', 'SOC_percent', 'nn'))
    data_json = json.dumps(data, default=str)
    return render(request, "visualisation_graphique_nun.html", {'data_json': data_json})

def GrapheDesDonnees_ndeux(request):
    data = list(Ndeux.objects.values('date','Energy_today_collected_kWh','Energy_today_consummed_kWh'))
    data_json = json.dumps(data, default=str)  # Convertir QuerySet en JSON
    return render(request, "visualisation_graphique_ndeux.html", {'data_json': data_json})

def GrapheDesDonnees_ntrois(request):
    data = list(Ntrois.objects.values('date', 'Total_Energy_today_collected_kWh', 'Total_Energy_today_consummed_kWh'))
    data_json = json.dumps(data, default=str)
    return render(request, "visualisation_graphique_ntrois.html", {'data_json': data_json})

def GrapheDesDonnees_nquatre(request):
    data = list(Nquatre.objects.values('date', 'Cell_voltage_min_V', 'Cell_voltage_max_V', 'Cell_voltage_bypass_V', 'nonnommeeun', 'nonnommeedeux', 'nonnommeetrois'))
    data_json = json.dumps(data, default=str)
    return render(request, "visualisation_graphique_nquatre.html", {'data_json': data_json})

def GrapheDesDonnees_ncinq(request):
    data = list(Ncinq.objects.values('date', 'Cell_voltage_lowest_V', 'Low_nr', 'Cell_voltage_highest_V', 'High_nr', 'Sbyteun', 'Sbytedeux'))
    data_json = json.dumps(data, default=str)
    return render(request, "visualisation_graphique_ncinq.html", {'data_json': data_json})

def GrapheDesDonnees_nsix(request):
    data = list(Nsix.objects.values('date','Tmp_lowest_deg',  'Tmp_highest_deg', 'Min_charg_temp_deg', 'Max_temp_deg'))
    data_json = json.dumps(data, default=str)  # Convertir QuerySet en JSON
    return render(request, "visualisation_graphique_nsix.html", {'data_json': data_json})
```

Views.py


Accueil
Tableau des données
Graphiques
Nzero
Nun
Ndeux
Ntrois
Nquatre
Ncinq
Nsix
Période :
-- Choisir une période --
jj/mm/aaaa --:--
jj/mm/aaaa --:--
Filtrer
Tension
Tension (V)
Cliquez-glissez pour zoomer
Tension (V)
100
95
90
19 févr.
06:00
12:00
18:00
20 févr.
Date et Heure
Highcharts.com
Courant Entrant
Courant Entrant (A)
Cliquez-glissez pour zoomer
Courant Entrant (A)
30
20
10
0
19 févr.
06:00
12:00
18:00
20 févr.
Date et Heure
Highcharts.com
Courant Sortant
Courant Sortant (A)
Cliquez-glissez pour zoomer
Courant Sortant (A)
6
4
2
0
Courant Batterie
Courant Batterie (A)
Cliquez-glissez pour zoomer
Courant Batterie (A)
30
20
10
0
-10

Données Batterie sur HiveMQ



mqtt_client.py 4 ×

ubuntu > dev > monProjetDjango > GestionAdmin > services > mqtt_client.py > ...

```python
BMS_TOPICS = [
    "bms/N0",
    "bms/N1",
    "bms/N2",
    "bms/N3",
    "bms/N4",
    "bms/N5",
    "bms/N6",
    # bms/cell/C5 (N7) absent du models.py → non géré
]


@shared_task
def process_bms_message(topic, payload_str):
    try:
        data = json.loads(payload_str)
        ts = datetime.fromisoformat(data['timestamp'])
        if ts.tzinfo is None:
            ts = timezone.make_aware(ts)


        from GestionExploitant.models import Nzero, Nun, Ndeux, Ntrois, Nquatre, Ncinq, Nsix

        if topic == "bms/N0":
            Nzero.objects.create(
                date        = ts,
                total_volt  = data.get('total_voltage_V', 0),
                currentin   = data.get('current_in_A', 0),
                currentout  = data.get('current_out_A', 0),
                current_bat = data.get('current_bat_A', 0),
            )

        elif topic == "bms/N1":
            Nun.objects.create(
                date                  = ts,
                Energy_stored_Wh      = data.get('Energy_stored_Wh', 0),
                Battery_capacity_kWh = data.get('Battery_capacity_kWh', 0),
                SOC_percent           = data.get('SOC_percent', 0),
                nn                    = data.get('unused', 0),  # champ non nommé dans le payload
            )

        elif topic == "bms/N2":
            Ndeux.objects.create(
                date                        = ts,
```

start_mqtt.py 1 ×

ubuntu > dev > monProjetDjango > GestionAdmin > management > commands > start_mqtt.py > ...

```python

from django.core.management.base import BaseCommand
from GestionAdmin.services.mqtt_client import MQTTClient

class Command(BaseCommand):
    help = 'Démarre le client MQTT'

    def handle(self, *args, **options):
        self.stdout.write(self.style.SUCCESS('Démarrage du client MQTT...'))
        client = MQTTClient()
        client.connect()

        # Gardez le processus en vie
        try:
            while True:
                pass
        except KeyboardInterrupt:
            client.disconnect()
            self.stdout.write(self.style.SUCCESS('Client MQTT arrêté'))

```

```
MQTT_BROKER_HOST = ''
MQTT_BROKER_PORT = 8883
MQTT_USERNAME = ''
MQTT_PASSWORD = ''
MQTT_TOPIC =''
MQTT_KEEP_ALIVE= 60
MQTT_CA_CERT="/home/ubuntu/dev/monProjetDjango/cert/isrgrootx1.pem"
MQTT_TLS_INSECURE = False
```

```
(envdjango) ubuntu@vps-62f78ac3:~/dev/monProjetDjango$ python manage.py start_mqtt
 Démarrage du client MQTT...
```

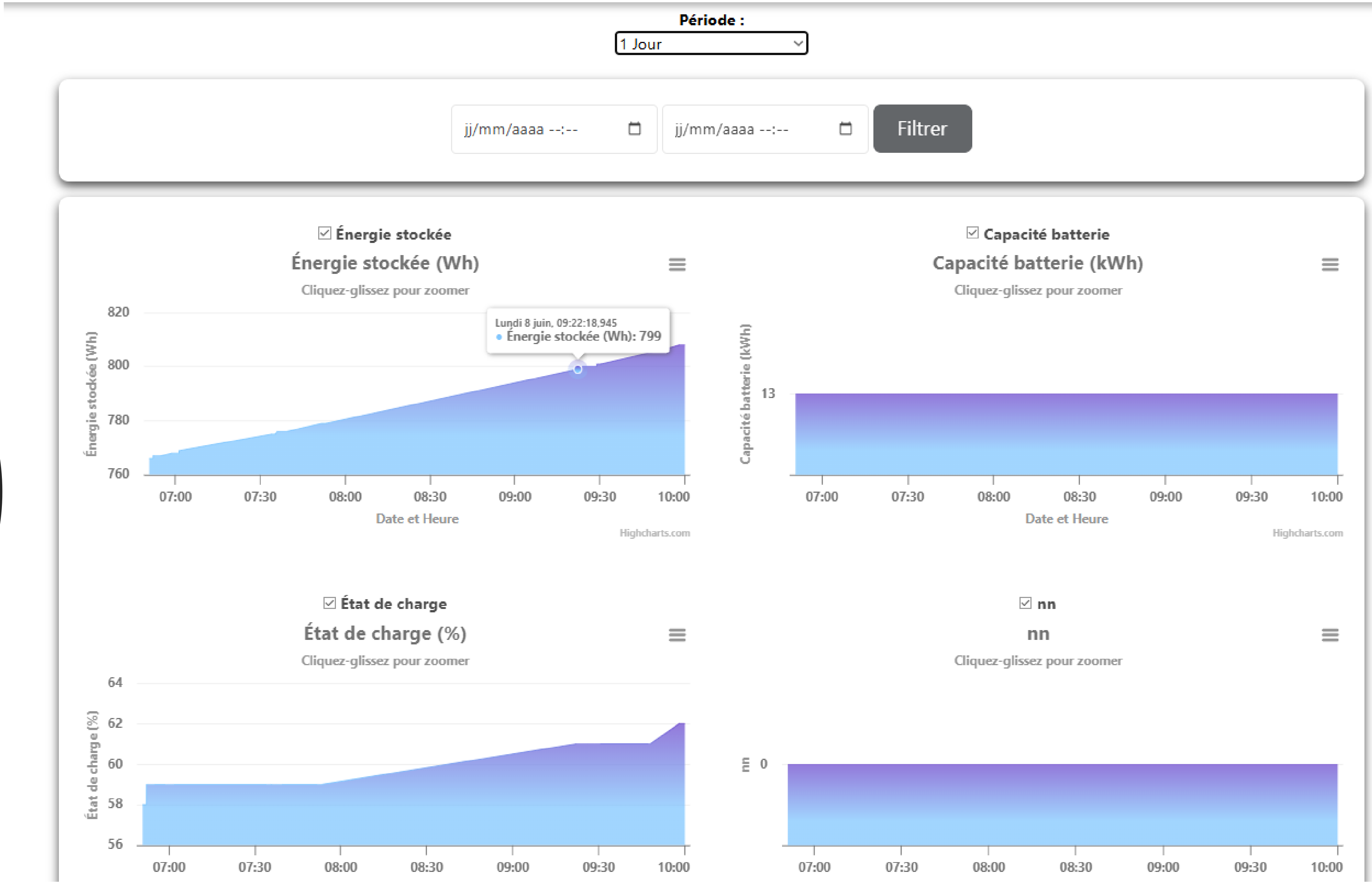
```
[2026-06-08 10:47:19,011: INFO/MainProcess] Connected to redis://localhost:6379/0
[2026-06-08 10:47:19,011: WARNING/MainProcess] /home/ubuntu/dev/envdjango/lib/python3.12/
g will no longer determine
whether broker connection retries are made during startup in Celery 6.0 and above.
If you wish to retain the existing behavior for retrying connections on startup,
you should set broker_connection_retry_on_startup to True.
  warnings.warn(

[2026-06-08 10:47:19,012: INFO/MainProcess] mingle: searching for neighbors
[2026-06-08 10:47:20,018: INFO/MainProcess] mingle: all alone
[2026-06-08 10:47:20,027: INFO/MainProcess] celery@vps-62f78ac3 ready.
```

- Configuration settings.py
- Démarrage de mqtt_client.py
- Démarrage de celery

Données récupérées sur Django

```
[2026-06-08 14:00:09,407: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[5884d228-f647-4929-adf0-9974e785cf23] received
[2026-06-08 14:00:09,422: INFO/ForkPoolWorker-4] BMS enregistré — topic: bms/N6 | ts: 2026-06-08 14:00:00.542380+04:00
[2026-06-08 14:00:09,423: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[5884d228-f647-4929-adf0-9974e785cf23] succeeded in 0.016127551905810833s: None
[2026-06-08 14:00:09,931: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[edb2122a-aa50-4310-84d3-26a24a64c490] received
[2026-06-08 14:00:09,948: INFO/ForkPoolWorker-4] BMS enregistré — topic: bms/N0 | ts: 2026-06-08 14:00:01.511685+04:00
[2026-06-08 14:00:09,948: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[edb2122a-aa50-4310-84d3-26a24a64c490] succeeded in 0.01632172986865043 6s: None
[2026-06-08 14:00:10,213: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[7b254712-d29c-49db-bd1f-34159eba05a9] received
[2026-06-08 14:00:10,230: INFO/ForkPoolWorker-4] BMS enregistré — topic: bms/N2 | ts: 2026-06-08 14:00:01.514042+04:00
[2026-06-08 14:00:10,230: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[7b254712-d29c-49db-bd1f-34159eba05a9] succeeded in 0.0162708000279963s: None
[2026-06-08 14:00:10,244: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[e7bfdac6-6ace-4d06-947d-86529735e139] received
[2026-06-08 14:00:10,260: INFO/ForkPoolWorker-4] BMS enregistré — topic: bms/N6 | ts: 2026-06-08 14:00:01.520412+04:00
[2026-06-08 14:00:10,261: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[e7bfdac6-6ace-4d06-947d-86529735e139] succeeded in 0.015957661904394627s: None
[2026-06-08 14:00:10,271: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[6e0f76b7-784a-4810-b5c2-ae9e88f3c3b6] received
[2026-06-08 14:00:10,286: INFO/ForkPoolWorker-4] BMS enregistré — topic: bms/N4 | ts: 2026-06-08 14:00:01.517316+04:00
[2026-06-08 14:00:10,287: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[6e0f76b7-784a-4810-b5c2-ae9e88f3c3b6] succeeded in 0.015664461068809032s: None
[2026-06-08 14:00:10,292: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[a45b75d8-01a5-4864-b640-d6a28527433a] received
[2026-06-08 14:00:10,308: INFO/ForkPoolWorker-4] BMS enregistré — topic: bms/N1 | ts: 2026-06-08 14:00:01.513023+04:00
[2026-06-08 14:00:10,309: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[a45b75d8-01a5-4864-b640-d6a28527433a] succeeded in 0.0163339939899742 6s: None
[2026-06-08 14:00:10,324: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[cb96559b-0059-47b3-bf4d-b141c9a2fcc5] received
[2026-06-08 14:00:10,340: INFO/ForkPoolWorker-4] BMS enregistré — topic: bms/N3 | ts: 2026-06-08 14:00:01.514998+04:00
[2026-06-08 14:00:10,340: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[cb96559b-0059-47b3-bf4d-b141c9a2fcc5] succeeded in 0.015993801876902 58s: None
[2026-06-08 14:00:10,366: INFO/MainProcess] Task GestionAdmin.services.mqtt_client.process_bms_message[c9f5d2a6-0395-4ee6-8b0f-16e5f0294287] received
[2026-06-08 14:00:10,382: INFO/ForkPoolWorker-4] BMS enregistré — topic: bms/N5 | ts: 2026-06-08 14:00:01.519155+04:00
[2026-06-08 14:00:10,382: INFO/ForkPoolWorker-4] Task GestionAdmin.services.mqtt_client.process_bms_message[c9f5d2a6-0395-4ee6-8b0f-16e5f0294287] succeeded in 0.015906693879514933s: None
```

Données Batterie sur le site



Sécurisation du serveur Django

OVH

- Serveur hébergé chez OVH

iptables / UFW

- Filtrage des connexions réseau.
- Autorisation uniquement des services nécessaires.
- Blocage des ports inutilisés.

Fail2ban

- Surveillance des tentatives de connexion.
- Détection des échecs répétés d'authentification.
- Bannissement automatique des adresses IP suspectes.

PV de Recette

N° test	Description du test	Résultat attendu	Résultat obtenu	Conforme
1	Connexion SSH	Accès au serveur	Connexion réussie	☒ Oui ☐ Non
2	Restauration BDD locale	Import de la sauvegarde	BDD accessible	☒ Oui ☐ Non
3	Chargement Highcharts local	Affichage sans CDN	Graphe affiché	☒ Oui ☐ Non
4	Ajout des données N+0 à N+6	Afficher sur le site avec graphe correspondant	Le site affiche chaque donnée avec leur graphe correspondant	☒ Oui ☐ Non
5	Connexion MQTT (HiveMQ) à Django	La connexion fonctionne avec envoie des données	La connexion n'a pas été établie	☐ Oui ☒ Non

Conclusion

Objectifs atteints

Optimisation restante

Compétence acquise

Suite du projet